FOUR EQUIVALENT CHARACTERIZATIONS OF FLAT MORPHISMS OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is the standard property-defines-property-morphisms template with locality of flatness as the input property

Editorial note. The label lemma-flat-morphisms names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the definition of flat morphisms of formal algebraic spaces (definition-flat) via etale covers and flat arrows of WAdm^Noeth
- flatness is a local property of arrows of WAdm^Noeth (lemma-flat-axioms of the same chapter)
- the Formal Spaces lemma that a local property of arrows defines a property of morphisms (formal-spaces-lemma-property-defines-property-morphisms) in its Noetherian variant (formal-spaces-remark-variant-Noetherian)

Source reference. Original reference 00HB: sources/stacks-restricted/pinned-original-source.tex, source lines 3023–3023. Prerequisite pointer invoked at proof line(s) 3079; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0AIM: sources/stacks-chow/tags.txt, source lines 13270–13270. Prerequisite pointer invoked at proof line(s) 3061; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ANG: sources/stacks-chow/tags.txt, source lines 13439–13439. Prerequisite pointer invoked at proof line(s) 3085; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0ANI: sources/stacks-chow/tags.txt, source lines 13441–13441. Prerequisite pointer invoked at proof line(s) 3089; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC9: sources/stacks-restricted/pinned-original-source.tex, source lines 2890–2890. Prerequisite pointer invoked at proof line(s)

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3082; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally Noetherian formal algebraic spaces over S. The following are equivalent*

(1) *f is flat,*
(2) *for every commutative diagram*

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

with U and V affine formal algebraic spaces, $U \to X$ and $V \to Y$ representable by algebraic spaces and étale, the morphism $U \to V$ corresponds to a flat map in $WAdm^{Noeth}$,
(3) *there exists a covering $\{Y_j \to Y\}$ as in Formal Spaces, Definition 0AIM and for each j a covering $\{X_{ji} \to Y_j \times_Y X\}$ as in Formal Spaces, Definition 0AIM such that each $X_{ji} \to Y_j$ corresponds to a flat map in $WAdm^{Noeth}$, and*
(4) *there exist a covering $\{X_i \to X\}$ as in Formal Spaces, Definition 0AIM and for each i a factorization $X_i \to Y_i \to Y$ where Y_i is an affine formal algebraic space, $Y_i \to Y$ is representable by algebraic spaces and étale, and $X_i \to Y_i$ corresponds to a flat map in $WAdm^{Noeth}$.*

Proof. The equivalence of (1) and (2) is Definition 00HB. The equivalence of (2), (3), and (4) follows from the fact that being flat is a local property of arrows of WAdm^{Noeth} by Lemma 0GC9 and an application of the variant of Formal Spaces, Lemma 0ANG for morphisms between locally Noetherian algebraic spaces mentioned in Formal Spaces, Remark 0ANI. □